STRATEGIC PLANNING REPORT

Project-Title: Data-Driven Inventory Management and Delivery Optimization for an E-Commerce Logistics Company

Project Area: Logistics Data Analysis
Internship/Program: Yuva Intern
Submitted by: Sankari D
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GitHub Project Link: https://github.com/sankari-d27/Week1_ecommerce-logistics-kpi-analysis

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Project Title

Data-Driven Inventory Management and Delivery Optimization for an E-Commerce Logistics Company

Overall Scenario

An e-commerce company receives online orders from customers across different cities. The company operates multiple warehouses and manages thousands of products.

The company wants to use historical e-commerce data to improve:

- Inventory availability
- Stock management
- Demand forecasting
- Warehouse allocation
- Delivery performance
- Shipping costs
- Customer satisfaction

Project Title: Week_1 Strategic Planning and Data Exploration for E-Commerce Logistics

1.1. Introduction

E-commerce logistics plays an important role in customer satisfaction, delivery reliability, and operational efficiency. Delays in shipping, inefficient order fulfilment, and high transportation costs can negatively affect business performance.

This project aims to analyse e-commerce logistics data using Python to evaluate delivery performance, calculate key performance indicators, and identify potential operational inefficiencies.

1.2. Business Scenario

The company maintains historical records of customer orders, products, warehouses, inventory, shipping, and deliveries.

The company wants to answer the following questions:

- Which products have the highest demand?
- Which products are slow-moving?
- Which products frequently experience stockouts?
- Which warehouses have excess or insufficient inventory?
- What factors affect delivery time?
- Which shipping methods are expensive?
- Which locations experience frequent delivery delays?
- How can inventory and delivery operations be optimized?

1.3. Project Objectives

This project focuses on analysing e-commerce logistics performance using key performance indicators (KPIs) such as delivery time, on-time delivery rate, order fulfilment rate, shipping cost, and delivery delays.

1. Understand the structure and quality of the logistics dataset.
2. Identify relevant logistics performance indicators.
3. Calculate delivery time, on-time delivery rate, and delivery delay rate.
4. Analyse order and shipping performance using Python.
5. Develop visualizations to communicate findings.
6. Provide data-driven recommendations for improving logistics operations.

Tools and Technologies

1. Python for data analysis.

2. Pandas and NumPy for data cleaning and calculations.
3. Matplotlib and Seaborn for data visualization.
4. Jupyter Notebook or Google Colab for implementation.
5. GitHub for version control and project documentation.

1.4. Key Performance Indicators

KPI 1 — Inventory Turnover

Inventory Turnover = Cost of Goods Sold / Average Inventory

This measures how efficiently inventory is being sold.

KPI 2 — Stockout Rate

Stockout Rate = Stockout Events / Total Demand Events × 100

A lower stockout rate is preferred.

KPI 3 — On-Time Delivery Rate

On-Time Delivery Rate = On-Time Deliveries / Total Deliveries × 100

KPI 4 — Average Delivery Time

average_delivery_time = df["Delivery_Days"].mean()

print(average_delivery_time)

KPI 5 — Average Shipping Cost

average_shipping_cost = df["Shipping_Cost"].mean()

print(average_shipping_cost)

KPI 6 — Order Fulfilment Rate

Fulfilment Rate = Successfully Fulfilled Orders / Total Orders × 100

Proposed Dataset

The dataset will contain variables such as:

Revenue Trend Over Time

Variable	Description
Order_ID	Unique order identifier
Order_Date	Date of order
Product_ID	Product identifier
Product_Category	Product category

Variable	Description
Quantity	Quantity ordered
Unit_Price	Price per unit
Warehouse	Warehouse location
Customer_City	Customer location
Shipping_Mode	Shipping method
Shipping_Cost	Shipping cost
Inventory_Level	Available inventory
Reorder_Level	Reorder threshold
Delivery_Days	Delivery duration
Order_Status	Order status
Customer_Rating	Customer rating

1.5. Data Science Techniques

The following techniques will be used:

- Exploratory Data Analysis: Understand sales, inventory, and delivery patterns.
- Regression: Predict product demand or delivery time.
- Clustering: Group products according to demand patterns.
- Classification: Predict whether an order may be delayed.
- Time-series analysis: Identify demand trends.
- Optimization: Determine inventory and warehouse allocation strategies.

1.6. Strategic Roadmap

E-Commerce Data

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Data Collection

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Data Cleaning

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Data Preprocessing

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Exploratory Data Analysis

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Inventory Analysis

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Delivery Analysis

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Demand Forecasting

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Delivery Prediction

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Inventory Optimization

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Delivery Optimization

↓

Business Recommendations

1.7. Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
print("Libraries imported successfully!")
```

Output: Libraries imported successfully!

Load dataset

```
import pandas as pd
df = pd.read_csv("ecommerce_logistics.csv")
print(df.head()) //Display data
```

Output:

Order_ID	Order_Date	Customer_ID	Product_Category	Order_Value\
ORD1001	2026-01-02	CUST001	Electronics	45000
ORD1002	2026-01-03	CUST002	Fashion	2500
ORD1003	2026-01-04	CUST003	Home & Kitchen	7800
ORD1004	2026-01-05	CUST004	Beauty	1500
ORD1005…	2026-01-06	CUST005	Electronics	32000

Shipping_Cost	Warehouse	Delivery_Partner	Delivery_Days	Expected_Days\
650	Bangalore	Delivery	3	4
180	Mumbai	BlueDart	2	3
320	Delhi	Ecom Express	5	4
120	Bangalore	Delivery	2	3
550	Hyderabad	BlueDart	4	4

Order_Status	Customer_Rating	Return_Flag
Delivered	5	No
Delivered	4	No
Delivered	3	No
Delivered	5	No
Delivered	4	No

Display data

```
print(df.head())
```

Dataset information

```
import pandas as pd
print(df.info()) //Dataset information
print(df.describe()) // Statistical summary
```

Output: Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Order_ID	30 non-null	Str
1	Order_Date	30 non-null	Str
2	Customer_ID	30 non-null	Str
3	Product_Category	30 non-null	Str
4	Order_Value	30 non-null	int64
5	Shipping_Cost	30 non-null	int64
6	Warehouse	30 non-null	Str
7	Delivery_Partner	30 non-null	str
8	Delivery_Days	30 non-null	int64
9	Expected_Days	30 non-null	int64
10	Order_Status	30 non-null	Str
11	Customer_Rating	30 non-null	int64
12	Return_Flag	30 non-null	Str

dtypes: int64(5), str(8)

memory usage: 3.2 KB

	Order_Value	Shipping_Cost	Delivery_Days	Expected_Days
Count	30.000000	30.000000	30.000000	30.000000
Mean	12346.666667	306.833333	3.566667	3.633333

Std	14150.441283	179.954416	1.501340	0.964305

25%	3.000000
50%	4.000000
75%	5.000000
Max	**5.000000**

Statistical summary

```
print(df.describe())
```

Check missing values

```
print(df.isnull().sum())
```

Average delivery time

```
import pandas as pd
average_delivery_time = df["Delivery_Days"].mean()
print("Average Delivery Time:", average_delivery_time)
```

Output: Average Delivery Time: 3.566666666666667

Average shipping cost

```
import pandas as pd
average_shipping_cost = df["Shipping_Cost"].mean()
print("Average Shipping Cost:", average_shipping_cost)
```

Output: Average Shipping Cost: 306.8333333333333

#Order Fulfilment Rate:

```
import pandas as pd
total_orders = len(df)
fulfilled_orders = len(df[df["Order_Status"] == "Delivered"])
fulfilment_rate = (fulfilled_orders / total_orders) * 100
print("Order Fulfilment Rate:", fulfilment_rate)
```

Output: Order Fulfilment Rate: 100.0

#Stockout Rate:

```
import pandas as pd
stockout_events = len(df[df["Inventory_Level"] <= df["Reorder_Level"]])
```

```
stockout_rate = (stockout_events / len(df)) * 100
print("Stockout Rate:", stockout_rate)
```

Output: Stockout Rate: 63.33333333333333

```
import pandas as pd
import matplotlib.pyplot as plt
# 1. Load dataset
df = pd.read_csv("ecommerce_logistics.csv")
# 2. Clean column names
df.columns = df.columns.str.strip()
# 3. Convert required columns to numeric
for col in ["Delivery_Days", "Expected_Days", "Shipping_Cost"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")
# 4. Validate required columns
required = ["Order_ID", "Delivery_Partner", "Delivery_Days", "Expected_Days", "Shipping_Cost"]
missing = [col for col in required if col not in df.columns]
if missing:
    raise ValueError(f"Missing columns: {missing}")
# 5. Keep rows with valid delivery information
valid_df = df.dropna(
    subset=["Order_ID", "Delivery_Partner", "Delivery_Days", "Expected_Days"]).copy()
# 6. Calculate on-time status
valid_df["On_Time"] = ( valid_df["Delivery_Days"] <= valid_df["Expected_Days"])
```

```
# 7. Summarize partner performance
partner_performance = (
    valid_df.groupby("Delivery_Partner")
    .agg(
        Total_Orders=("Order_ID", "nunique"),
        On_Time_Rate=("On_Time", "mean"),
        Avg_Shipping_Cost=("Shipping_Cost", "mean")))
```

```
# 8. Convert rate to percentage
partner_performance["On_Time_Rate"] *= 100
# 9. Display results
display(partner_performance.round(2))
# 10. Plot delivery partner performance
partner_performance["On_Time_Rate"].sort_values().plot(
    kind="barh",
    color="steelblue",
    figsize=(9, 5)
)
plt.title("On-Time Delivery Rate by Delivery Partner")
plt.xlabel("On-Time Delivery Rate (%)")
plt.ylabel("Delivery Partner")
plt.tight_layout()
plt.show()
```

1.8. Expected Outcome

The analysis is expected to provide a clearer understanding of delivery reliability, shipping performance, and potential logistics bottlenecks. Actual findings will be documented after the data has been processed and evaluated.

1.9 Conclusion

Week 1 established the foundation for the E-commerce Logistics KPI Analysis project by defining the objectives, identifying relevant KPIs, and planning the Python-based analysis.

Following the feedback received, the report will be strengthened with detailed implementation, numerical examples, visualizations, and evidence-based findings. These improvements will provide a stronger foundation for the subsequent project work.